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NPS/003/013 – Technical Specification for Fault Passage Indicators for use with Ground Mounted 11 & 20kV Switchgear

1. Purpose

This specification details the technical requirements for Fault Passage Indication on ground mounted plant and switchgear on the CE Electric UK distribution networks.

This document supersedes the following document, all copies of which shall be destroyed.

Ref	Version	Title
NPS/003/013	1.0	Specification for Fault Passage Indicators for use with Ground Mounted 11 & 20kV Switchgear

2. Scope

This specification covers Fault Passage Indicators (FPI) and associated Current Transformers (CTs) for use with ground mounted plant and switchgear on the CE Electric UK 11 & 20kV distribution networks.

3. Technical Requirements

3.1 Overview

The requirement is for FPIs and CTs that can be used on the CE Electric UK 11 & 20kV distribution network.

FPIs and CTs will be used on new installations. Inbuilt CTs, installed and wired by the switchgear manufacturers, shall be utilised. FPI relays for use on new installations shall be compatible with CTs inherent to the switchgear.

FPIs and CTs will be used on existing installations to upgrade, repair or replace existing FPI facilities. The FPI and/or CT might be required to be compatible with switchgear remote control facilities on site and CTs inherent to the switchgear.

The FPI and CTs arrangement on new switchgear shall be capable of detecting Earth Fault current, and/or Phase-to-Phase current, and/or Three-Phase Fault current.

In existing switchgear situations where it is not possible to fit CTs suitable for the detection of Phase-to-Phase and Three-Phase fault currents, the FPI and CT arrangement shall be capable of detecting Earth Fault currents.

The FPI may be required to interact with switchgear remote control facilities on site. The FPI shall have volt free contacts for interaction with remote control schemes, etc.

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The equipment shall comply with all relevant and current: IEC International Standards, British Standard Specifications or equivalent Euro-Norms, and Energy Networks Association Technical Specifications (ENATS), except where varied by this standard.

3.2 Technical Specification

3.2.1 Overall

The FPI and associated CT(s) shall meet the requirements of ENATS 48-2 for type A & B FPIs except where varied by this specification.

The FPI shall not be battery powered.

FPIs and associated equipment shall be capable of withstanding, without any detriment to their operation and without suffering damage or degradation, the effects of CT outputs resulting from passage of Earth Fault current up to 5kA for 3 seconds, or phase to phase fault current up to 20kA for 3 seconds.

The Earth Fault CT primary to secondary turns ratio shall be 60/1.

The Phase to Phase Fault CT primary to secondary turns ratio shall be 400/1

3.2.2 Fault Detection

The unit shall detect and indicate the passage of earth fault current for currents of 40A or above.

The unit shall not operate for the passage of earth fault currents for currents of 20A or below; this is to avoid misleading indication resulting from the detection of stray current related to earth faults on other parts of the network.

Where a sensitive earth fault setting of less than 30A is provided; operation of the FPI shall be delayed for a period of between 4 and 8 seconds to prevent operation of the FPI as a result of faults on other parts of the system.

Where appropriate, the unit shall detect and indicate the passage of phase to phase and three-phase fault currents for currents of 600 amps or above and shall be stable and not operate below this value.

3.2.3 Fault Indication

The local indication of fault current passage shall be clear and unambiguous and shall distinguish between earth fault and overcurrent (phase-to-phase and three-phase) faults.

Where a means of indication employs a flashing light, then the flash rate shall not be less than once every 3 seconds.

Where earth fault and overcurrent faults are distinguished by a different flash-rate of the same indicator in the same colour, the flash rate of overcurrent faults shall be the faster and differ from that of earth faults by a factor of 2 or more. In situations where both earth and overcurrent faults have been detected, the indication shall be at the highest flash rate.

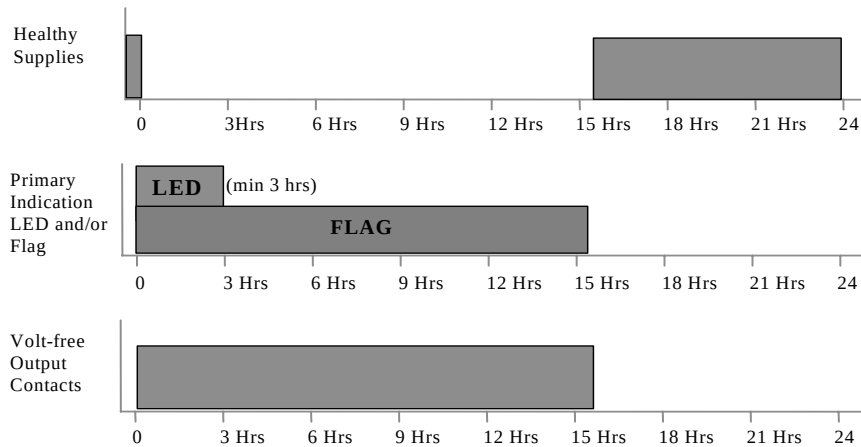
The indication shall display until the unit is reset, or for a minimum of 3 hours if the unit is not reset within this period.

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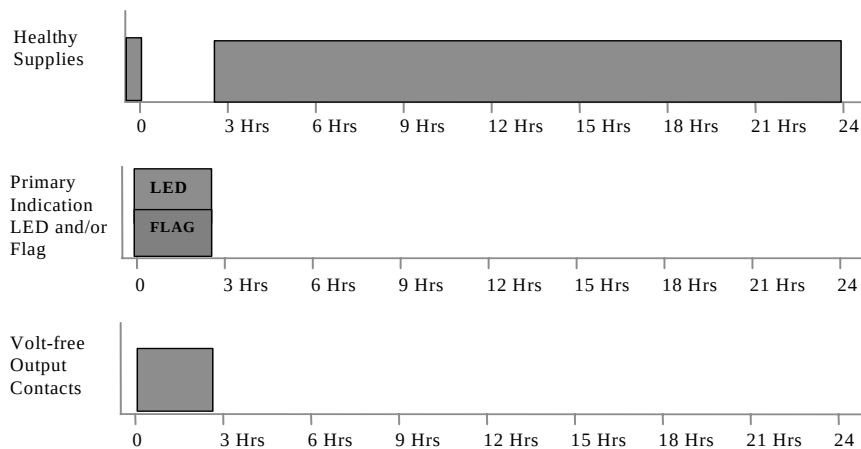
Normally open, volt free, contacts, corresponding to the state of the FPI, shall be provided. These volt free contacts shall remain latched until the unit is reset by either: restoration of supplies, remote reset command or manual reset, but in any case not by loss of internal energy.

The diagrams below indicate how the indication and volt free output contacts are required to operate.

Minimum requirements for the operation of indication and output contacts for supplies restored in excess of 3 hours



Minimum requirements for the operation of indication and output contacts for supplies restored within 3 hours



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3.2.4 Resetting

Situation 1 – For use where a 230V reset supply is available

The FPI relay shall be capable of being reset automatically by restoration of a 230V AC input supply, manually locally or via remote SCADA command.

Upon receipt of a 230V signal for a continuous period of 10 seconds the FPI shall automatically reset the local indication and the volt free contacts. This 10 second delay is to avoid false resets resulting from short term circuit re-energisation during fault switching.

Facilities for manually resetting the FPI shall be on the front panel of the relay.

Following operation of the FPI relay the volt free output contacts shall remain closed until the unit is reset by one of the three methods specified above.

Following a reset, the unit shall be ready and capable to carry out a full detection and indication sequence again within 30 seconds of receipt of the reset command or signal.

Situation 2 – For use where a 230V reset supply is not available

The FPI relay shall be capable of: self reset (after a selectable time delay, with a minimum time of 3hrs), being reset manually locally or being reset via remote SCADA command.

Facilities for manually resetting the FPI shall be on the front panel of the relay.

Following operation of the FPI relay the volt free output contacts shall remain closed until the unit is reset by one of the three methods specified above.

Following a reset, the unit shall be ready and capable to carry out a full detection and indication sequence again within 30 seconds of receipt of the reset command or signal.

3.2.5 Testing Facilities

The relay shall provide means to allow testing of the indication(s) without the need for tools, equipment or instruments.

The relay shall provide facilities to allow testing of the volt free contact(s).

The FPI and the CT shall allow in situ testing of the installed arrangement by primary injection through the CT and shall also allow in situ testing of the installed arrangement by secondary injection of the FPI.

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3.2.6 Environmental Type Testing

General

The material of the housing shall be resistant to the degrading effects of ultraviolet radiation.

Where the housing is of a sealed design it shall be protected to a minimum level of IP65 in accordance with IEC 60529.

Where the housing is of a ventilated design it shall be protected to a minimum level of IP55 in accordance with IEC 60529, any condensation within the housing shall free drain from the housing and the circuit board and components shall be coated with a moisture barrier.

A single FPI relay unit shall be used for the environmental tests listed below, which shall be carried out in the sequence listed below. The unit shall be:

Attached to the test equipment by using the bracket arrangement that will be used to mount the unit in a substation

In the quiescent state during the test

Recovered at standard atmospheric conditions.

Following testing the unit shall be examined for signs of mechanical damage and the correct operation for detection of fault conditions verified.

Temperature

The FPI shall function fully and correctly within the range -25°C to +40°C.

The FPI shall meet the requirements of IEC 60068-2-1 severity class -25°C 16 hours.

The FPI shall meet the requirements of IEC 60068-2-2 severity class +40°C 16 hours.

The test shall take into account solar radiation, at 1000W/m², in accordance with 60068-2-5. This solar gain shall be applied to the test, at least, during the last 8 hours of test.

Impact

The FPI shall meet the requirements of IEC 60068-2-62; 2J test with 3 impacts being applied to each side of the body, to the face of the unit and to the top of the unit.

Relative Humidity

The FPI shall operate satisfactorily with a relative humidity up to 100%.

The FPI shall meet the requirements of IEC 60068-2-30 with severity class 40°C 56 cycles.

In the case of vented designs care shall be taken to ensure that when external moisture is removed from the unit any moisture that has condensed within the unit is neither shaken from the unit or provided with a wick allowing it to be withdrawn from the unit.

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Salt Mist

The FPI shall meet the requirements of IEC 60068-2-52 - 3 day test

The unit shall be recovered and the exterior washed and dried before examination for material degradation or corrosion and functionality checks. The condition of the interior of the unit and internal components shall also be checked for evidence of corrosion or material degradation.

Electromagnetic Emissions

The FPI shall meet the requirements of IEC 61000-6-3 for Radiated Electric Field Emissions.

Electromagnetic Immunity

The FPI shall meet the requirements of IEC 61000-6-2 for:

Power Frequency magnetic field

Immunity to Radio Frequency Electromagnetic Fields,

Immunity to Electrostatic Discharge.

The FPI shall meet the requirements of EN 60255-22-4: class A (4kV 5/50 nSec. 2.5kHz repetitive)

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4. References

4.1 External Documentation

Reference	Title
ENATS 48-2	FPIs for 6.6kV & 11kV Underground and Overhead Distribution Systems
IEC 60068	Environmental Testing
IEC 60529	Degrees of Protection Provided by Enclosures
IEC 61000	Electromagnetic Compatibility (EMC)
IEC 60255-22-4	Relays - Electrical Fast Transient/Burst Immunity

4.2 Internal documentation

Reference	Title
NONE	

5. Definitions

Term	Definition
NONE	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant & Switchgear Standards Manager	D Blackledge	15/02/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	26/02/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	03/03/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	20/03/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	03/03/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	11/03/08

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Appendix 1 - Addendum to Supplier

New Installations v Existing Installations

The specification states that:

The FPI and CT arrangement shall be capable of detecting Earth Fault current, and/or Phase-to-Phase current, and/or Three-Phase Fault current

It is accepted that on the majority of existing FPIs, many of which have a single phase balance CT fitted around a three phase cable, it may not be practical to convert these to also detect phase fault currents. E.g. where an existing site employs a 3-core cable entering into a compound filled cable box.

Variants of FPI and of CTs

The specification states that:

"FPIs and CTs will be used on new installations, where they may be required to interact with switchgear remote control facilities on site and the FPI may be required to be compatible with CTs inherent to the switchgear.

FPIs and CTs will be used on existing installations to upgrade, repair or replace existing FPI facilities. The FPI and/or CT may be required to be compatible with either: switchgear remote control facilities on site and/or CTs inherent to the switchgear."

Where practicable manufactures shall offer a single design of FPI that whilst meeting all the requirements of the main specification incorporates the facility to connect to a range of CT assemblies. The combined arrangement shall satisfy all the scenarios listed below. Where this is not practicable CE Electric UK will consider multiple FPI designs.

- In conjunction with CTs (which may be inherent to the switchgear) that are existing, are fitted with the FPI, or are fitted later.
- and
- In conjunction with CTs in an arrangement that can distinguish between earth fault and phase to phase faults.
- and
- In substations with, or without, a 230V reset/charging supply.

Where practicable manufactures shall offer a single design of CT that whilst meeting all the requirements of the main specification offers the facility to connect to a range of different cable arrangements. Typical scenarios are listed below. Where this is not practicable CE Electric UK will consider multiple CT arrangements.

- In conjunction with FPIs that are existing, are fitted with the CT, or are fitted later.
- and
- In situations requiring the fitting of CTs to existing cables
- and
- In situations that require the detection of, and distinguishing between, earth fault and phase to phase faults.

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11 & 20kV Cables

CE Electric UK three core cable has an outside diameter of between 35mm and 100mm

CE Electric UK single core cable has an outside diameter of between 61mm and 81mm when in trefoil formation

CE Electric UK single core cable has an outside diameter of between 28mm and 49mm when not in trefoil formation

3 Core cable enters the cable box before the cores are separated.

Single core cable is usually laid in the ground in trefoil formation and the cores then either:

(a) Remain in trefoil until after they enter the cable box

or

(b) Are separated before entering the cable box, in flat formation at approx. 110mm centres and the trefoil separation point is typically 500mm below the cable box. At this distance below the cable box it is likely that any CT fitted around the trefoil portion of the cable is likely to be in, or close to the bottom of the cable trench; this is not be an acceptable arrangement.

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Appendix 2 – Compliance Self Declaration Table

To be completed by suppliers.

Where possible; details/information to be entered in right hand column,
as opposed to 'Yes' or 'No' entries.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

NPS/003/013 - FPIs	
3.1 Overview	
Suitable for use on new installations, including utilising switchgear manufactures inbuilt CTs	
Existing Installations	
Earth Faults & Phase to Phase faults	
Resettable: manually, auto, or remotely via 230V supply	
Volt-free contacts	
ENATS, BS & IEC compliance	
3.2 Tech Spec	
3.2.1 Overall	
ENATS 48-2	
Not battery powered	
Fault current withstand: Phase-Earth = 5kA for 3 secs Phase - Phase = 20kA for 3 secs	
CT ratio: Earth fault = 60:1 Phase - Phase fault = 400:1	
3.2.2 Fault Detection	
Phase - Earth will detect @ 40A or above	
Phase - Earth will stable up to 20A	
Sensitive Earth Fault (<30A) has 4 to 8s delay	
Phase - Phase fault will detect at 600A, or above, is stable and will not operate below 600A,	
3.2.3 Fault Indication	
Local indication is clear and unambiguous	

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Local indication flash rate maximum 3 secs	
Local indication for at least 3 Hrs, or until reset	
Normally open volt free contacts provided	
Normally open volt free contacts remain latched until reset	
Indication and volt free contacts compliance with illustrated characteristics in diagrams	
3.2.4 Resetting (Situation 1)	
Reset automatically (230V with 10sec delay)	
Reset manually locally (on front panel)	
Reset remotely via Scada - <i>state reset signal characteristics</i>	
Upon any reset - indication stops and volt free contacts reset	
After reset; FPI is ready again <= 30 secs - state recovery time	
3.2.4 Resetting (Situation 2)	
Reset manually locally (on front panel)	
Reset remotely via Scada - <i>state reset signal characteristics</i>	
Upon any reset - indication stops and volt free contacts reset	
After reset; FPI is ready again <= 30 secs - state recovery time	
3.2.5 Testing	
Testing does not require special tools, etc	
Testing of volt free contacts is provided	
FPI allows secondary injection testing	
CT/FPI allows primary injection testing	
3.2.6 Environmental Testing	
3.2.6 General	
UV resistant	
If sealed, is IP65	
If ventilated is free draining & components coated	
If ventilated is IP55	
3.2.6 Temperature	
OK for -25C to +40C	
IEC 600368-2-1 -25C for 16 Hrs	
IEC 600368-2-1 +40C for 16 Hrs (inc solar gain 1000W/m ² for, at least, last 8 hours)	
3.2.6 Impact	
IEC60068-2-62 2J x 3 impacts to: each side, face and top	

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3.2.6 Relative Humidity	
up to 100%	
IEC60068-2-30 40C 56 cycles	
3.2.6 Salt Mist -	
IEC60068-2-52 3 day test	
3.2.6 Electromagnetic Emissions	
IEC61000-6-2 – radiated electric field emissions	
4.6.9 Electromagnetic Immunity	
IEC60068-2-2 - power freq electromag field	
IEC60068-2-2 - radio freq electromag field	
IEC60068-2-2 - immunity to electrostatic discharge	
EN 60255-22-4 class A (4kV 5/50 nSec 2.5kHz repetitive)	
Appendix 1 - addendum to suppliers	
Single CT variant will suit all installations <i>- if not please detail minimum variations</i>	
Single FPI relay will suit all installations <i>- if not please detail minimum variations</i>	
CTs are compatible with:	
3 core cable - diameter 35-100mm	
1 core cable, in trefoil - diameter 61-81mm	
1 core cable - diameter 28-49mm	
Cable box, 3 hole gland plate 110mm centres	

Manufacturer:

Product Reference:

Name:

Signature:

Date: